The New School
Parsons School of Design
Design and Technology
PSAM 2230 A
Device Art
CRN : 15895
Thursday, 7-10pm, 66 Fifth Ave, N818

Fall 2024 Course Syllabus

Complete Syllabus Overview

Instructor Information

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Available for consultation by appointment. Wednesday preferred.

Course Description

Device Art is a platform for students to invent new works of art that do not distinguish the object from a tool, the mechanism from the concept; through playfulness these devices are caught between definitions of art, design, and engineering. In this course devices are the artwork themselves; the mechanisms of the piece become part of the concept behind the work and the very essence of "making" and "play" is embedded in each of the pieces.

Device Art can break the mold of art commercialization, propelled by the new forms of manufacturing and the ease of prototyping as production tools become more accessible. These artworks are easily replicable and can become commercialized, and even display features of playful utility for everyday life. Students will learn new methods for prototyping by combining new digital fabrication techniques as well as commonly used fabrication processes including repurposing readymades. They will explore their own artistic concepts and develop a deeper understanding of how their ideas can be reproduced.

Device Art will challenge students to rethink their ideas of what art can become, and where the threshold lies between playfulness, utility, and conceptual inquiry. They will develop basic electronics and physical computing skills, such as microcontroller programming and circuitry design. They will engage in fabrication experiments to rethink and envision new devices that can facilitate invention from an artistic perspective, gaining new skills in device design, computing media, and device fabrication. They will use physical computing methods to expand on the field of open source hardware, learn interaction design methods and rapid prototyping tools for 3D printing. This course will provide a platform to enhance the design process of material experimentation and propel functional and accessible models of device art.

Open to: All university undergraduate degree students. Some seats have been reserved for BFA Design & Technology majors.

Learning Outcomes

By the successful completion of this course, students will be able to:

1. Integrate skills in physical computing and fabrication to create unique, multi-sensory interactive works.
2. Gain a practical understanding of electronics and physical computing, including voltage, current, resistance, power, and more.
3. How to prototype and break down an idea into a circuit, components, and code. 4. Use of Arduino and Arduino IDE to program microcontroller driven projects 5. Incorporation of sensors, actuators (motors), and OLED displays into basic electronic circuits 6. Safely work with electricity, batteries, and power supplies for your electronic projects. 7. Build a final interactive project that integrates varying media and sensory themes in a creative practice of their own choosing.
8. Basic understanding of 3D modeling (Fusion 360) and FDM 3D printing.
9. Basic understanding of Laser cutting and Adobe Illustrator.

Assessable Tasks

- **Midterm assignment**, Due 11/8/24, create a device art piece, give an in-class presentation, submit a short write up with a work statement and description of your creative process. - **Final assignment**, due 12/13/24, create a device art piece, give an in-class presentation, submit a short write up with a work statement and description of your creative process. - **Attendance**: attend every class or complete alternative assignments for missed classes. - **Assignments**: completion of video and other reading assignments.

Evaluation and Final Grade Calculation

Attendance: 25%
Homework assignments: 25%
Midterm Project: 25%
Final Project: 25%

Meaningful Participation and Attendance

Class participation is an essential part of class and includes: keeping up with reading, assignments, projects, contributing meaningfully to class discussions, active participation in group work, and attending sessions regularly and on time.

The attendance guidelines were developed to encourage students' success in all aspects of their academic programs. Full participation is essential to the successful completion of coursework and enhances the quality of the educational experience for all, particularly in courses where group work is integral. Thus, Parsons promotes high levels of attendance. Students are expected to attend classes regularly and promptly and in compliance with the standards stated in this course syllabus.

While attendance is just one aspect of meaningful participation, absence from a significant portion of class time may prevent the successful attainment of course objectives. A significant portion of class time is generally defined as the equivalent of three weeks, or 20%, of class time. Lateness or early departure from class may be recorded as one full absence. Students may be asked to withdraw from a course if habitual absence or tardiness has a negative impact on the class environment.

I will assess each student's performance according to all of the assessment criteria specified in this syllabus in determining your final grade.

Final Grade	**Description**	**Percentage (Undergraduate Only)**	**GPA Analog**
A	Work of exceptional quality, which often goes beyond stated goals of the course	95% - 100%	4.0
A-	Work of very high quality	90% - <95%	3.7
B+	Work of high quality; Higher than average abilities	87% - <90%	3.3
B	Very good work; satisfies the goals of the course	83% - <87%	3.0
B-	Average work; average understanding of course material	80% - <83%	2.7
C+	Below average work; understanding of course material	77% - <80%	2.3
C	Acceptable work; passable	73% - <77%	2.0
C-	Passable work but below good academic standing	70% - <73%	1.7
D	Well below good academic standing	60% - <70%	1.0
F	Failure, no credit	0% - <60%	0

***Note:** *Grades of D are not used in graduate level courses.*

Course Readings, Materials, and Technology Requirements

Readings

LinkedinLearning (Formerly Lynda.com): video tutorials available for free through the New School Libraries. Access here: https://guides.library.newschool.edu/LinkedInLearning

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Materials, Supplies, and Technology

Hardware:

Arduino Starter Kit: you may use one from a previous course, or substitute the kit for the following parts. ~$110

- Arduino Uno or equivalent
- Breadboard, jumper wires, resistors
- USB cable rated for data+power
- DC motor , stepper motor

Digital Multimeter: preferred
Personal Computer: preferably a laptop to bring to class

Software:

Arduino IDE : free
Adobe Illustrator: free with student Adobe CC access
Fusion 360: free for personal use, student version free as well
Ultimaker Cura or other 3D printing slicer: free

Use of Generative Artificial Intelligence (AI) Tools.

You may use any AI tools you like for this course. We will discuss how to use tools such as ChatGPT effectively for troubleshooting code and responsibly for augmenting your learning.

Course Outline

WEEK 1	Aug 29	**Introduction to Device Art**	Assignment: Buy Arduino Kit, Complete Course Survey, Watch Intro to Electronics video
WEEK 2	Sept 5	**Sensors + Switches**	Assignment: Watch Intro to Arduino video
WEEK 3	Sept 12	**Microcontrollers**	Sensor Assignment, FastLED Library Video
WEEK 4	Sept 19	**Outputs and Interactions, LEDs**	Assignment: LED Assignment Bring a simple electronic gadget next class
WEEK 5	Sept 26	**Readymades and Re-enclosures**	Watch Sound Video
WEEK 6	Oct 3	**Sound**	Watch Fusion360 Video

WEEK 7 Oct 10 **Digital Fabrication Methods I** Work on Midterm project WEEK 8 Oct 17 **Midterm Project Presentations** Assignment:

			Reading for next class XXX Due:
WEEK 9	Oct 24	**Digital Fabrication Methods II**	Watch Motor Video
WEEK 10	Oct 31	**Actuation : Things that move**	TBD
WEEK 11	Nov 7	**Enclosures + Miniaturization**	TBD
WEEK 12	Nov 14	**Advanced Circuits, Final Project announcement**	Schedule time to discuss the final project. Watch OLED Video
WEEK 13	Nov 21	**OLED Displays + Guest Lecture**	TBD
	Nov 28	**THANKSGIVING BREAK**	
WEEK 14	Dec 5	***Final Project Help***	Finish Final Project
WEEK 15	Dec 12	**Final Presentation and Critiques**	N/A

University, College, School, and Program Policies

Academic Honesty

Compromising your academic integrity may lead to serious consequences, including (but not limited to) one or more of the following: failure of the assignment, failure of the course, academic warning, disciplinary probation, suspension from the university, or dismissal from the university.

Students are responsible for understanding the University's policy on academic honesty and integrity and must make use of proper citations of sources for writing papers, creating, presenting, and performing their work, taking examinations, and doing research. It is the responsibility of students to learn the procedures specific to their discipline for correctly and appropriately differentiating their own work from that of others. The full text of the policy, including adjudication procedures, is found here.

Resources regarding what plagiarism is and how to avoid it can be found on the Learning Center's website.

The New School views "academic honesty and integrity" as the duty of every member of an academic community to claim authorship for his or her own work and only for that work, and to recognize the contributions of others accurately and completely. This obligation is fundamental to the integrity of intellectual debate, and creative and academic pursuits. Academic honesty and integrity includes accurate use of quotations, as well as appropriate and explicit citation of sources in instances of paraphrasing and describing ideas, or reporting on research findings or any aspect of the work of others (including that of faculty members and other students). Academic dishonesty results from infractions of this “accurate use”. The standards of academic honesty and integrity, and citation of sources, apply to all forms of academic work, including submissions of drafts of final papers or projects. All members of the University community are expected to conduct themselves in accord with the standards of academic honesty and integrity.

TNS Student Disability Services

If you are a student with a disability/disabled student, or believe you might have a disability that requires accommodations, please head to the SDS website, and complete the Self ID form. Then, head to Starfish and find a time to meet with Nick Faranda, at a time of mutual convenience. If you have any questions or concerns, please contact the Student Disability Services (SDS) at studentdisability@newschool.edu, or 212-229-5626.

Student Course Ratings (Course Evaluations)

During the last two weeks of the semester, students are asked to provide feedback for each of their courses through an online survey. They cannot view grades until providing feedback or officially declining to do so. Course evaluations are a vital space where students can speak about the learning experience. It is an important process which provides valuable data about the successful delivery and support of a course or topic to both the faculty and administrators. Instructors rely on course rating surveys for feedback on the course and teaching methods, so they can understand what aspects of the class are most successful in teaching students, and what aspects might be improved or changed in future. Without this information, it can be difficult for an instructor to reflect upon and improve teaching methods and course design. In addition, program/department chairs and other administrators review course surveys. Instructions are available online here.

Additional University-wide Policies

- Intellectual Property Rights
- TNS Grading Policies
- Title IX Policy

A comprehensive overview of University policies may be found under Policies: A to Z. Students are also encouraged to consult the Academic Catalog.

Course-specific Policies

Responsibility

Students are responsible for all assignments, even if they are absent. Late papers, failure to complete the readings assigned for class discussion, and lack of preparedness for in-class discussions and presentations will significantly impact your successful completion of this course.

Canvas

Use of Canvas may be an important resource for this class. Students should check it for announcements before coming to class each week.

Electronic Devices

The use of electronic devices (phones, tablets, laptops, cameras, etc.) is permitted when the device is being used in relation to the course's work. All other uses are prohibited in the classroom and devices should be turned off before class starts.

Resources

The university provides many resources to help students achieve academic and artistic excellence. These resources include:

● The University Libraries
The New School Libraries provide access to a vast array of print and electronic resources as well as personal research consultations, classroom instruction, and spaces for study and collaboration.

● Archives & Special Collections and Digital Collections
The New School Archives and Special Collections holds a wide array of collections in many different formats that may be useful in your academic, artistic, and personal projects, including paper and digital records, audiovisual material, artist's books, zines, and records related to the histories of all divisions of the University. Archivists are available to help with your research and to offer guidance for locating resources specific to your topic. Contact archivist@newschool.edu to get started.

● The University Learning Center
For assistance with coursework during the semester, I encourage you to schedule free tutoring sessions at the University Learning Center (ULC). Individual appointments in Writing, Software, Computer Programming, Oral Presentations, Math, Time Management and ADHD Coaching are available from 7am-midnight Monday-Friday and 12-5pm on Saturdays. Online appointments are scheduled via WCONLINE and in person sessions or last minute virtual walk-ins can be requested by emailing learningcenter@newschool.edu. In person sessions are held at 66 W. 12th St. on the 6th floor. The ULC also offers weekly and biweekly sessions. For a complete list of services and general information, please visit the ULC webpage.

● Making Center

The Making Center is a constellation of shops, labs, and open workspaces that are situated across the New School to help students express their ideas in a variety of materials and methods. We have resources to help support woodworking, metalworking, ceramics and pottery work, photography and film, textiles, printmaking, 3D printing, manual and CNC machining, and more. A staff of technicians and student workers provide expertise and maintain the different shops and labs. Safety is a primary concern, so each area has policies for access, training, and etiquette with which students and faculty should be familiar. Many areas require specific orientations or trainings before access is granted.

● The New School Food Assistance includes food assistance and additional resources for New School students.

● Health and Wellness includes additional services and support available to New School students. 7

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